Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "round trip" and "signal" to explain your answer. Sentence starters are optional.

round trip: travel from start to a reflector and back

signal: a disturbance that carries information

Doppler shift: a frequency change associated with relative motion

01 Warm-up: bring back an earlier idea

An EM wave in vacuum has frequency **5×10^{9} Hz**.
1. Use c=**3.00×10^{8} m/s**.
2. **Find wavelength** and **name the spectral region**.

Word help: vacuum = space with negligible matter

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

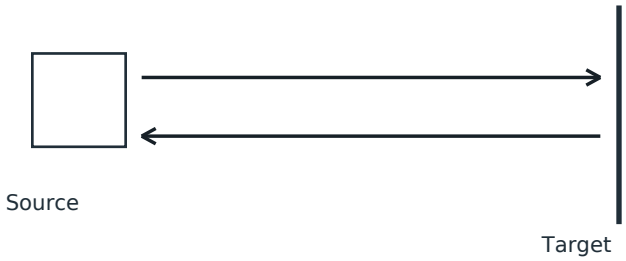
02 Worked example: explain the prediction

Reference: range = c Δt/**2**; c = **3.0×10^{8} m/s**.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after **2 microseconds**. With c=**3×10^{8} m/s**, range=c Δt/**2**=**300 m**. The measured time includes the outward and return paths.

Your turn: Draw the round trip and **label the conversion 2 microseconds=2×10^{-6} s**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Draw outgoing** and returning radar paths.
2. **Which measurement** helps range: echo delay or frequency shift?
3. Which helps radial velocity?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: 1 2 3

04 Apply the physics

A stationary radar receives an echo after **0.040 s**. Use c = **3 × 10⁸ m/s** to **find range**. Separately, its wavelength is **0.10 m** and Doppler-shift magnitude is **400 Hz**. Use $|v_r| = |\Delta f|\lambda/2$ to **find radial speed**. **Label which measurement supports each** result.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? **Sketch another possible motion**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Add a velocity perpendicular to the line of sight.



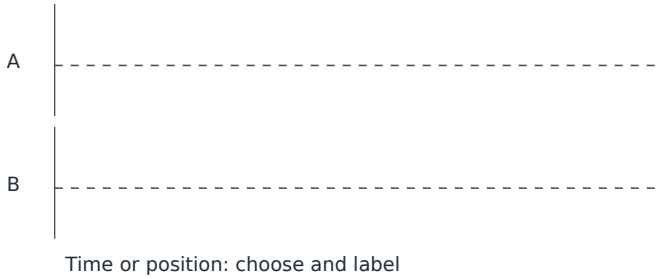
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave travels **12 m** in **0.4 s** at frequency **60 Hz**. **Find speed** and wavelength.
Label a wave sketch to **support the calculation or comparison**.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.

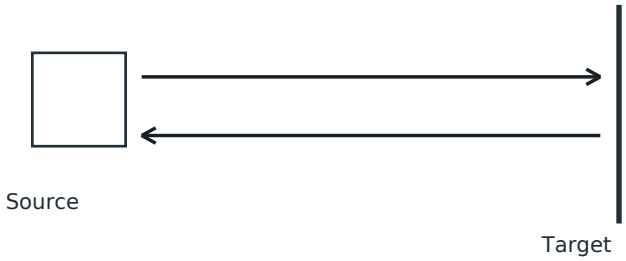


Marzano check - circle: 1 2 3

07 Exit check

Two targets at the same range have different Doppler shifts. **Explain why they can have** same time-delay but different motions.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Radar range changes from **1500** to **1440 m** over **0.5 s**.
1. **Find average radial velocity taking** outward positive.
2. **Explain why total speed may** be larger.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3